Representation Collapse and Recovery in Lightweight Encoders: The Critical Role of Embedding Dimensionality

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***Abstract*—We study representation bottlenecks in lightweight encoders under increasing data complexity. Rather than relying on large pretrained backbones, we conduct controlled experiments to isolate the effects of parameter budget, model scaling, and embedding dimensionality under constrained settings.**

Across MNIST, CIFAR-10, and ImageNet-based experiments, we observe a consistent progression. In low-complexity regimes, lightweight encoders maintain strong class separability even under very small parameter budgets. In contrast, under high-complexity settings such as ImageNet-1000, simply increasing model size or depth yields only marginal improvement. However, increasing embedding dimensionality produces substantial gains, revealing a three-phase behavior of collapse, recovery, and saturation.

These findings suggest that, in lightweight representation learning, embedding dimensionality plays a critical role in representation quality and may provide greater benefit than naive model scaling in complex data regimes.

***Index Terms*—Embedding dimensionality, lightweight models, representation learning, metric learning, bottleneck analysis**

I. Introduction

Recent progress in deep learning has largely been driven by scale. Larger models, deeper networks, and large-scale pretraining have consistently improved performance across many tasks. However, many real-world systems do not enjoy unlimited computational resources. Edge devices, embedded systems, mobile platforms, and low-power deployments often require lightweight encoders with strict parameter budgets.

This raises a fundamental but underexplored question:

> *What fundamentally limits representation quality in lightweight encoders as data complexity increases?*

A common answer would be model size. In many modern settings, when performance degrades, the default solution is to scale the network by increasing depth, width, or parameter count. Yet under constrained regimes, this strategy may not be feasible. More importantly, model size alone may not explain representational failure.

In this work, we argue that embedding dimensionality plays a more direct role than model scaling in determining representation quality for lightweight encoders. Our experiments suggest a structured progression: (1) **collapse**, where increasing data complexity causes representation quality to break down; (2) **recovery**, where increasing embedding dimensionality alone restores separability; and (3) **saturation**, where further dimensional increase yields diminishing returns.

The central contribution of this paper is not the introduction of a new architecture, but the empirical isolation of embedding dimensionality as a key control variable in constrained representation learning.

Our contributions are summarized as follows:

- We show that lightweight encoders remain effective in low-complexity regimes such as MNIST and CIFAR-10.
- We demonstrate that naive model scaling provides only limited gains in ImageNet-1000.
- We identify dimensionality-induced recovery, where embedding dimensionality alone improves separability.
- We provide evidence of saturation beyond a critical embedding regime, suggesting diminishing returns after sufficient representational capacity is reached.

II. Related Work

A. Metric Learning and Embedding Quality

Metric learning has long focused on constructing embedding spaces in which intra-class samples are close and inter-class samples are separated. Early approaches such as contrastive learning and triplet loss demonstrated that discriminative embeddings can significantly improve downstream recognition tasks. FaceNet [1] is one of the most influential examples, showing how structured embeddings can be used for large-scale recognition.

More recently, supervised contrastive learning [2] has shown that embedding quality can be directly optimized through label-aware contrastive objectives. These approaches emphasize the importance of embedding structure, but they typically assume sufficiently powerful backbones and do not explicitly study the effect of embedding dimensionality under strict parameter constraints.

B. Scaling and Model Capacity

Scaling laws [3] suggest that performance improves predictably with model size, dataset size, and computation. In

computer vision, models such as ResNet [4] and EfficientNet [5] exemplify the benefits of scaling architectural capacity.

However, this dominant scaling perspective does not fully answer the lightweight setting considered here. When model capacity is constrained by design, increasing parameters may be insufficient or infeasible. Our study therefore complements model scaling literature by asking whether embedding dimensionality can serve as a more efficient route to improved representation.

C. Large-Scale Representation Learning

Modern representation learning methods such as SimCLR [6], BYOL [7], and CLIP [8] highlight the power of large-scale feature learning. These methods typically rely on strong encoders, long training schedules, large batches, or massive datasets. By contrast, our setting is intentionally constrained. We focus on what happens when the encoder is lightweight and representational capacity must be budgeted carefully.

D. Representation Bottlenecks

The broader representation learning literature has repeatedly suggested that compression, information flow, and internal bottlenecks play an important role in neural networks. Bengio et al. [9] and Tishby et al. [10] provide conceptual foundations for understanding why representational bottlenecks may limit downstream performance. Yet these ideas are rarely connected directly to embedding dimensionality in controlled lightweight settings.

Our work contributes to this direction by studying bottlenecks not only in terms of abstract information flow, but also in terms of concrete embedding dimensionality under fixed and small-capacity models.

III. Method

A. Problem Setting

Let x denote an input sample and let a lightweight encoder $f_\theta(\cdot)$ map the sample to an embedding vector:

$$z = f_\theta(x), \quad z \in \mathbb{R}^D \tag{1}$$

where D is the embedding dimensionality. Our objective is to analyze how representation quality changes as a function of data complexity, parameter budget, and embedding dimension.

The key constraint in this paper is that the encoder is intentionally lightweight. We do not rely on large pretrained backbones, nor do we frame the study as an architecture competition. Instead, we adopt controlled experiments to isolate which factors matter most under small-capacity regimes.

B. Representation Quality Metric

To quantify representation quality, we use the Gap Index:

$$G = \mathbb{E}[\mathrm{Dist}_{diff}] - \mathbb{E}[\mathrm{Dist}_{same}] \tag{2}$$

where Dist_{same} denotes the average distance between embeddings of samples from the same class and Dist_{diff} denotes the average distance between embeddings of samples from different classes.

A larger positive Gap indicates better class separation in the embedding space. A Gap close to zero suggests weak separation, while negative values indicate severe entanglement between classes.

C. Why Does Dimensionality Matter?

Increasing embedding dimensionality effectively increases the representational degrees of freedom available to the model.

In low-dimensional spaces, many classes are forced to share overlapping regions of the embedding space, which can lead to entanglement and collapse. As dimensionality increases, the model can allocate more independent directions for separating semantic structure.

From this perspective, embedding dimensionality affects the capacity of the representation space itself, rather than merely the feature extractor. This distinction is central to our analysis: even when encoder structure is fixed, changes in D can materially alter the separability of learned representations.

D. Interpretation of Collapse, Recovery, and Saturation

We use the following qualitative interpretation throughout the paper:

- **Collapse:** representation quality degrades toward zero or below zero as data complexity increases.
- **Recovery:** a substantial improvement in Gap occurs when embedding dimensionality is increased without changing encoder structure.
- **Saturation:** additional dimensionality beyond a sufficiently large regime yields limited marginal gains.

This framing does not claim a universal law for all models and datasets. Rather, it provides a descriptive vocabulary for the controlled lightweight regime studied in this work.

IV. Experiments

A. Complexity-Induced Collapse

We first examine how representation quality changes as data complexity increases. Figure 1 summarizes the degradation from low-complexity to high-complexity datasets.

Figure 1 shows a monotonic degradation in representation quality as we move from MNIST to CIFAR-10, ImageNet-10, ImageNet-100, and ImageNet-1000. In MNIST, lightweight encoders produce highly separable representations. In CIFAR-10, representation quality remains positive, though lower than MNIST. As complexity increases further, the Gap declines substantially and eventually becomes negative in the ImageNet-1000 setting.

This result suggests that lightweight encoders are not universally weak; rather, they fail selectively when the representational burden imposed by the dataset becomes too large. The collapse is therefore tied to data complexity rather than to the mere existence of a small model.

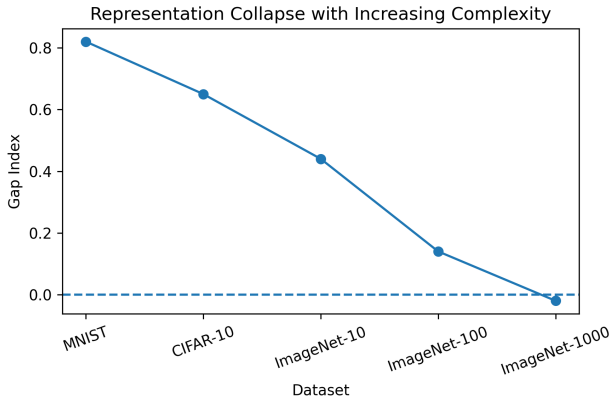

Fig. 1. Representation collapse with increasing complexity. The Gap decreases from MNIST and CIFAR-10 to ImageNet-100 and becomes negative in ImageNet-1000.

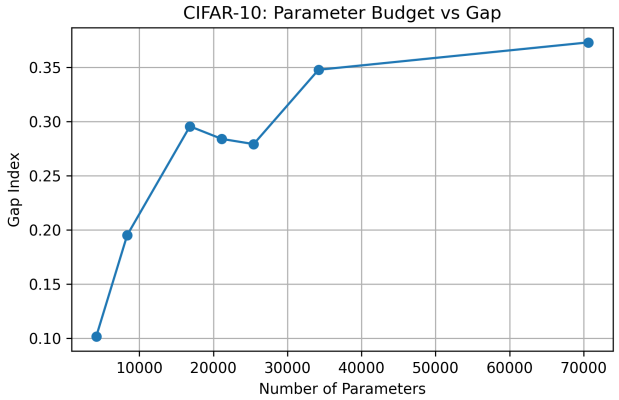

Fig. 2. CIFAR-10 parameter scaling. As the parameter budget increases, the Gap improves steadily, indicating that lightweight encoders remain effective in moderate-complexity settings.

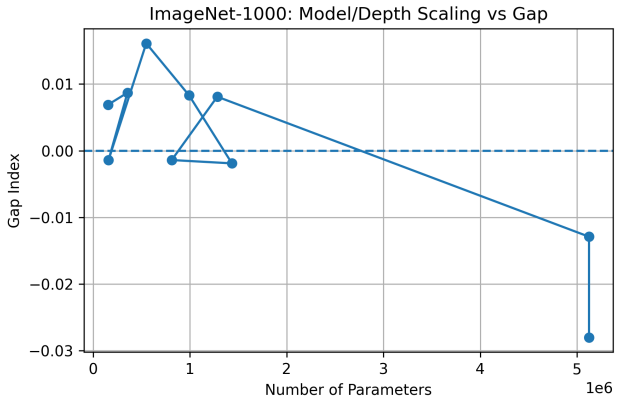

Fig. 3. ImageNet-1000 model/depth scaling vs. Gap. Increasing parameters or depth yields only limited gains, with most settings remaining near zero.

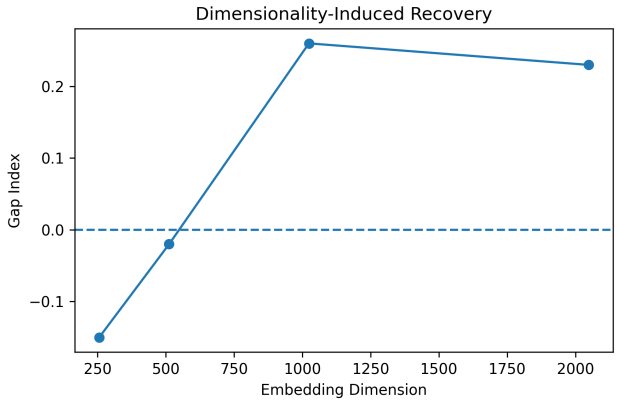

Fig. 4. Dimensionality-induced recovery. Increasing embedding dimensionality produces a substantial improvement in representation quality without modifying encoder structure.

B. Low-Complexity Regime: Parameter Scaling Works

We next focus on CIFAR-10, where parameter scaling behaves in a more conventional way.

Figure 2 shows that increasing the parameter budget on CIFAR-10 yields a consistent increase in Gap. This indicates that in moderate-complexity settings, additional capacity in the encoder directly translates into better representational separation.

The importance of this result is comparative. If lightweight encoders improve normally on CIFAR-10 but fail on ImageNet-1000, then the failure in the latter case cannot be dismissed as a generic weakness of small models. Instead, it points to a complexity-dependent bottleneck.

C. Failure of Model Scaling in ImageNet-1000

We then examine whether the same scaling behavior holds in the high-complexity regime.

Figure 3 demonstrates that increasing model parameters or depth does not substantially improve representation quality in ImageNet-1000. Even when the parameter budget increases dramatically, the Gap remains close to zero, and in several cases becomes negative.

This is a key negative result. If simply adding more parameters were sufficient, then the Gap should improve in the same way it does on CIFAR-10. Instead, the trend remains flat and unstable. This suggests that the bottleneck is not located purely in upstream feature extraction capacity.

In other words, naive model scaling provides only limited benefit in the high-complexity lightweight regime.

D. Dimensionality-Induced Recovery

We next ask whether the bottleneck lies in the embedding space itself.

Figure 4 reveals a striking effect. When embedding dimensionality is increased, representation quality improves dramatically, even though the encoder architecture itself remains unchanged.

This behavior differs qualitatively from the failure of model scaling. Increasing dimensionality does not merely produce a small incremental gain; rather, it restores a level of class separability that was absent in lower-dimensional settings.

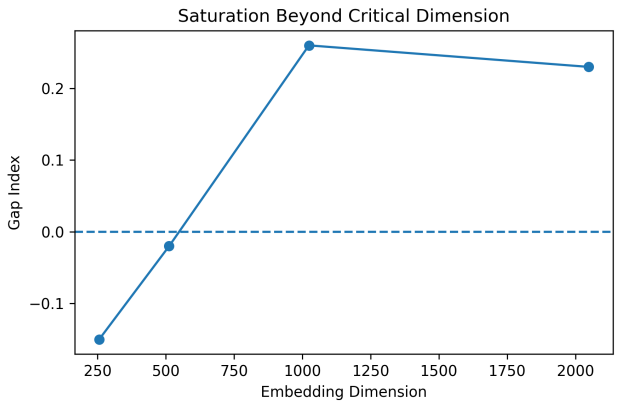

Fig. 5. Saturation beyond threshold. Performance improves sharply around the recovery regime and then exhibits diminishing returns.

TABLE I
ADDITIONAL VALIDATION UNDER SUPERVISED CONTRASTIVE LEARNING.

Dim	Gap	kNN
512	0.0731	0.0504
1024	0.0797	0.0640

This suggests that an important bottleneck in high-complexity lightweight representation learning lies not only in feature extraction, but also in representational bandwidth at the embedding layer.

E. Saturation Beyond the Critical Regime

Recovery alone is not the full story. The gains obtained by increasing dimensionality eventually begin to saturate.

As shown in Figure 5, once the embedding dimension becomes sufficiently large, the marginal benefit of further increases becomes limited. This suggests that the representation bottleneck is relieved after a critical dimensional regime is reached.

The implication is important: embedding dimensionality matters strongly, but only up to a point. After sufficient representational freedom is achieved, other factors may become the limiting source of performance.

F. Additional Validation

Finally, we include the additional 100-class validation experiment using supervised contrastive learning and k-NN evaluation.

Table I shows that increasing embedding dimensionality from 512 to 1024 improves both Gap and k-NN accuracy in the validation setting. Although the gain is smaller than in the main ImageNet-1000 experiment, the direction of the result is consistent. This supports the claim that embedding dimensionality contributes robustly to representation quality beyond a single experimental configuration.

V. RESULTS

Taken together, the experiments reveal a consistent three-phase structure.

First, as data complexity increases, lightweight encoders enter a collapse regime in which representation quality degrades sharply. Second, increasing embedding dimensionality induces recovery, restoring class separability without changing the encoder architecture. Third, beyond a sufficiently large embedding regime, gains begin to saturate.

This pattern suggests that representation quality in lightweight encoders is governed less by naive model size and more by the interaction between data complexity and embedding dimensionality.

VI. DISCUSSION

The experimental evidence supports the hypothesis that embedding dimensionality plays a critical role in lightweight representation learning.

We therefore introduce the notion of a critical dimensional regime D^*, defined not as a universal numeric constant, but as the range of embedding sizes in which meaningful separation first emerges for a given data complexity and encoder family.

The results suggest the following qualitative behavior:

- For $D < D^*$, embeddings are too compressed and representations collapse.
- Around $D \approx D^*$, increasing dimensionality yields strong recovery.
- For $D > D^*$, additional dimensionality yields diminishing returns.

This behavior resembles a phase transition in effective representational capacity. Although we do not claim a formal theorem, the empirical pattern is strong enough to motivate this interpretation.

More broadly, our findings suggest that lightweight model design should not focus exclusively on parameter efficiency or architectural compactness. Instead, embedding dimensionality should be treated as a first-class design variable, especially for complex data regimes.

VII. CONCLUSION

We studied representation bottlenecks in lightweight encoders under increasing data complexity. Our results reveal a three-phase behavior of collapse, recovery, and saturation.

Most importantly, we find that increasing embedding dimensionality provides a much more effective route to improved representation than naive model scaling in high-complexity settings.

These findings suggest that embedding dimensionality strongly affects representation capacity in lightweight encoders.

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